MATH210B Homework 7

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1 Problem 1

Let K/F be a field extension and let $\alpha, \beta \in K$. Assume α and β are algebraic over F, of respective degrees m and n.

Lemma 1.1. Let m' be the degree of α over $F(\beta)$. Then, β has degree $\frac{m'n}{m}$ over $F(\alpha)$.

Proof. Notice that $[F(\alpha, \beta) : F(\alpha)][F(\alpha) : F] = [F(\alpha, \beta) : F(\beta)][F(\beta) : F]$.

Since $[F(\alpha) : F] = m$ and $[F(\beta) : F] = n$ and $[F(\alpha, \beta) : F(\beta)] = m'$, the result immediately follows. The only lemma we need to continuously apply is that the degree of an element over a field is also the degree of the extension. $\square$

Lemma 1.2. If m and n are coprime, $[F(\alpha, \beta) : F] = mn$.

Proof. By the Tower Law, $[F(\alpha, \beta) : F] \leq [F(\alpha) : F][F(\beta) : F] = mn$.

Also notice that $m \mid [F(\alpha, \beta) : F]$ and $n \mid [F(\alpha, \beta) : F]$. Since m and n are coprime, $mn \leq [F(\alpha, \beta) : F]$.

We thus conclude the proof. $\square$

2 Problem 4

$y = \frac{x^2}{x-1}$ if and only if $y(x-1) = x^2 = 0$ if and only if $x^2 - yx + y = 0$, so x satisfies a quadratic polynomial over $F(y)$. Notice that x is not in $F(y)$ since y is irreducible in $F(y)$ and $y = \frac{x^2}{x-1}$ would contradict this. Thus, the degree of the extension is 2.

3 Problem 5

Let $f(x) = x^3 - 2 \in \mathbb{Q}[x]$.

Notice that f is irreducible since we can apply Eisenstein's with $p = 2$. Notice that 2 doesn't divide the monic term but divides the constant term. Moreover, 4 doesn't divide the constant term.

Let $w = e^{\frac{2\pi i}{3}} = \frac{-1}{2} + i\frac{\sqrt{3}}{2}$.

Notice that the roots of f are $\sqrt[3]{2}$, $\sqrt[3]{2}w$, $\sqrt[3]{2}w^2$.

Thus, if we add all roots of f, we're also adding $\sqrt{3}$.

Notice that $[\mathbb{Q}[\sqrt{3}] : \mathbb{Q}] = 2$ and $[\mathbb{Q}\sqrt[3]{2} : \mathbb{Q}] = 3$.

By Problem 1 on this homework, since 2 and 3 are relatively prime, we have that the degree of the extension is 6.

4 Problem 6

5 Problem 7

We present an example of a non-trivial field extension K/F with isomorphic fields.

Consider $F = \mathbb{C}[x_1, \ldots]$.

Now, consider $F(y)$. This is a non-trivial field extension since it's a transcendental field extension of F. However, both of these fields are isomorphic since we can map $x_2 \mapsto x_1, x_3 \mapsto x_2 \cdots$ and map $x_1 \mapsto y$.

However, finding a finite field extension would be better. Notice that the extension in Problem 4 satisifies this!

6 Problem 8

Lemma 6.1. Let K/F be a field extension and $\alpha \in K$. If $f(\alpha)$ is algebraic over F for some non-constant $f \in F[x]$, then α is algebraic over F.

Proof. If $f(\alpha)$ is algebraic over F, there exists some $g \in F[x]$ such that $g(f(\alpha)) = 0$. But then α is a root of $g \circ f$ and therefore algebraic over F. $\square$

7 Problem 9

Notice that α and β are the roots of $x^2 - (\alpha + \beta)(x) + \alpha\beta$ which is in $F(\alpha + \beta, \alpha\beta)$.

8 Problem 10

We present an infinite algebraic extension. Consider adding adjoining all pth roots of unity to $\mathbb{Q}$ for every prime p. This is an algebraic extension since every element satisfies $f(x) = x^p - 1$ for some $p > 0$ prime. Moreover, for a pth root of unity ζ, we have that $[\mathbb{Q}(\zeta) : \mathbb{Q}] = p - 1$. Thus, the field extension is infinite.